

EXHALE: A 1-D Multi-frequency Photoionization–Hydrodynamics Model of Escaping Exoplanet Atmospheres

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ABSTRACT

We present EXHALE, a one-dimensional, multi-frequency photoionization–hydrodynamics model for the XUV-driven escape of close-in exoplanet atmospheres and for the transit-absorption diagnostics that trace it. EXHALE evolves the compressible Euler equations for a single bulk fluid on a spherical grid with a Godunov finite-volume scheme (HLLC/Roe fluxes, PLM and WENO3 reconstruction) and relaxes to a stationary transonic wind, with an optional Jacobian-free Newton–Krylov (JFNK) solver that drives the discrete steady residual to machine-relevant levels. The thermal and ionization state is set by a coupled nonlinear system for hydrogen, helium, and the trace metals C, N, O, Mg, Ca, Na, and Fe, solved with an analytic-Jacobian Newton iteration and a MINPACK fallback, using Badnell radiative and dielectronic recombination, Voronov collisional ionization, and Kingdon & Ferland charge exchange with hydrogen. Cooling includes Ly α , recombination, free–free, and closed-form metal-line coefficients fitted to CHIANTI. EXHALE solves the metastable He 2³S population that produces the He I 10830 Å triplet, builds the non-LTE $n = 2$ hydrogen population for the Balmer lines with an in-line Ly α escape-probability radiative-transfer field, allows diffusive He/H (and He/metal) separation, and can adopt a lower-atmosphere base from a photochemical (VULCAN) column. We apply the model to HD 209458 b, HD 189733 b, WASP-52 b, and WASP-121 b and compare synthetic He 10830, H α , Ly α , and metal-line transit depths against published observations. The four Newton-converged models span mass-loss rates $\log_{10}(\dot{M}/\text{g s}^{-1}) \approx 9.0\text{--}13.2$, from the compact HD 189733 b wind to the strongly irradiated WASP-121 b, with thermosphere temperature maxima of $\sim 0.8\text{--}1.2 \times 10^4$ K. The synthetic depths range from sub-percent H α on the quieter targets to tens of percent in He 10830 and Ly α , and are compared line by line with published measurements; the model reproduces the order of magnitude of the observed He 10830, H α , and Ly α absorption on the two cooler targets, over-predicts the WASP-52 b depths, and on WASP-121 b saturates against the truncation of the modeled domain, motivating the composition and domain refinements discussed in the text.

Keywords: planets and satellites: atmospheres — hydrodynamics — radiative transfer — methods: numerical

1. INTRODUCTION

Close-in giant planets receive intense stellar X-ray and extreme-ultraviolet (XUV) irradiation. Photoionization deposits energy in the upper atmosphere, heats it to $\sim 10^4$ K, and drives a transonic hydrodynamic wind that carries mass away from the planet (E. N. Parker 1958; R. A. Murray-Clay et al. 2009). This escape sculpts the mass–radius distribution of the exoplanet population and can be observed directly through excess absorption in a handful of spectral tracers during transit.

Four diagnostics dominate the observational picture. The He I 10830 Å triplet arises from the metastable He 2³S level, which is populated by recombination of

He II and depopulated by photoionization, radiative decay, and collisions; because the mid-ultraviolet flux that photoionizes He 2³S and the XUV flux that produces He II depend on the host star, the 10830 line is a sensitive probe of both the wind structure and the stellar spectrum (A. Oklopčić & C. M. Hirata 2018; D. Yan et al. 2022). The hydrogen Balmer lines, chiefly H α , absorb from the $n = 2$ level, whose population is set primarily by Ly α radiative pumping and thus requires a resonant-line radiative-transfer treatment (D. Christie et al. 2013; C. Huang et al. 2017). Ly α itself is a strong but interstellar-absorbed tracer of neutral hydrogen. Finally, metal resonance lines (Mg II h&k, Ca II H&K, Na I D, and Fe II) probe the ionization and thermal structure

of the wind and, for the hottest planets, its metal content (C. Huang et al. 2023; T. T. Koskinen et al. 2022a).

A number of one-dimensional escape models have been built to interpret these tracers, spanning steady-state boundary-value solvers (R. A. Murray-Clay et al. 2009), time-relaxation photoionization-hydrodynamics codes (M. Salz et al. 2016; A. Caldiroli et al. 2021), grid-production schemes (D. Kubyshkina et al. 2018), and dedicated multispecies thermosphere models with excited-state microphysics (A. R. Taylor et al. 2025, 2026). **EXHALE** belongs to this last, diagnostic-oriented family. It began as a fork of the **ATES** photoionization hydrodynamics code (A. Caldiroli et al. 2021, 2022) and retains its Godunov hydro core and boundary/initial-condition skeleton, while adding a coupled trace-metal ionization network with self-consistent equation of state, CHIANTI metal-line cooling, the $\text{He } 2^3S$ metastable network, the non-LTE $n = 2$ hydrogen population with in-line $\text{Ly}\alpha$ radiative transfer, diffusive He/H separation, a direct steady-state (JFNK) solver, and a tiered coupling to the lower atmosphere. This paper documents the physics and numerics of **EXHALE** and applies it to four benchmark planets.

Section 2 describes the model. Section 3 contrasts **EXHALE** with **ATES** in both algorithm and physics. Section 4 presents the application to each planet (scaffold), and Section 5 summarizes.

2. METHOD

2.1. Hydrodynamics and numerical scheme

EXHALE solves the one-dimensional, spherically symmetric Euler equations for mass, radial momentum, and total energy of a single bulk fluid, with source terms for gravity, tidal (Roche) and centrifugal forces, photoionization heating, and radiative cooling. The discretization is a finite-volume Godunov method inherited from **ATES**: fluxes are computed with an approximate Riemann solver (HLLC by default; Roe and local Lax–Friedrichs are also available), the interface states are reconstructed either with a piecewise-linear method (PLM, MC-limited) or with a third-order essentially non-oscillatory scheme (ESWENO3), and time integration uses a strong-stability-preserving Runge–Kutta update at CFL number 0.6 (A. Caldiroli et al. 2021; M. Salz et al. 2016). The grid has 500 cells plus ghost cells on a mixed uniform–stretched mesh spanning the base radius to the outer boundary. The spacing of the uniform region is a runtime quantity, because it sets the number of cells per base density scale height $H = k_B T / (\mu g)$, and a base resolved by too few cells supports a stationary cell-to-cell entropy (contact) mode that the non-dissipative Riemann fluxes do not damp. The code re-

ports $H(T_{\text{eq}})/\Delta r$ at startup; on the default grid this is ≈ 27 cells for HD 189733 b against ≈ 102 for WASP-121 b, and the high-gravity, compact HD 189733 b is correspondingly the planet of the four that needs a refined base. In this respect **EXHALE** and the **PLUTO**–**CLOUDY** interface of M. Salz et al. (2016) are numerical near-twins, sharing the same reconstruction, Riemann solver, and time integrator.

The stationary transonic wind is reached in two complementary ways. The default is time relaxation, converged on the spatial flatness of the mass flux $\rho v r^2$ over the wind window,

$$du = \frac{\max_j |\rho v r^2|_j - \min_j |\rho v r^2|_j}{\min_j |\rho v r^2|_j} < du_{\text{th}}, \quad (1)$$

with a two-stage schedule that marches first with PLM (robust) and then with WENO3 (low-diffusion) to a final tolerance $du_{\text{th}} = 10^{-3}$. This mass-flux criterion has two known blind spots: it cannot detect a uniform level drift of an already-flat profile, and on the strongly driven planets it dips transiently below threshold long before the wind settles, so a premature snapshot can be reported as steady. **EXHALE** therefore adds (i) a level-stability gate on the windowed mean of $|\rho v r^2|$ and (ii) a diagnostic that evaluates the finite-volume steady residual $\mathbf{R}$ directly (mass, momentum, and heating-minus-cooling energy balance), which ranks candidate states independently of du and can serve as the convergence test.

For quantitative mass-loss rates **EXHALE** provides a direct steady-state solver that finds the root of the discrete residual, $F(Y) = 0$, where $Y = (\rho, \rho v, E)$ on the physical cells and the ionization equilibrium (with its heating and cooling) is eliminated locally inside every residual evaluation. The nonlinear system is solved by pseudo-transient-continuation Newton iteration with an inexact Jacobian-free Newton–Krylov (JFNK) inner solve: a banded finite-difference Jacobian, obtained from a graph-colored stencil and factored with LAPACK, preconditions a matrix-free GMRES whose Jacobian–vector products capture the weakly non-local radiation response. Robust convergence required a smooth ℓ_2 line-search merit, diagonal scaling across the several decades spanned by the base-to-wind density contrast, a differentiable base velocity valve, frozen WENO smoothness weights during the inner solve, and a non-monotone (Grippe) step acceptance. The solver reaches steady residual levels unattainable by time marching at a small fraction of the cost; a best-iterate/fail-fast guard falls back to plain marching if the Newton basin is not reached, so an unconverged state cannot be mistaken for the answer.

The lower boundary is a fixed reservoir. Density and temperature are held at the base radius (total $n \sim 10^{14} \text{ cm}^{-3}$ near a μbar pressure level, T at the planet’s equilibrium value), and the ghost-cell velocity uses a one-way valve $v_{\text{gh}} = \max(v_1, 0)$ that admits wind-feeding inflow but suppresses the base oscillations (“breathing”) that otherwise develop at steep base density gradients — the same valve adopted independently by [A. Caldiroli et al. \(2021\)](#) and [M. Salz et al. \(2016\)](#). **EXHALE** adds a dynamic base-pressure correction that pins the base temperature exactly through the equation of state, and, for the Newton solver, a smooth variant of the valve. The upper boundary is supersonic free outflow, imposed by zero-gradient or linear extrapolation at the outer (Roche-lobe) radius. Within physically realistic ranges the mass-loss rate is insensitive to the base density, temperature, and ionization fraction but scales steeply with the base radius, $\dot{M} \propto R_p^3$ ([R. A. Murray-Clay et al. 2009](#); [M. Salz et al. 2016](#)); **EXHALE** therefore treats the base radius (the pressure level it corresponds to) as the physically important lower-boundary quantity.

2.2. Photoionization, heating, and multi-frequency radiation

The stellar XUV field is attenuated radially and drives photoionization and photoelectric heating. **EXHALE** integrates each species’ energy-dependent photoionization cross section against the local stellar spectrum on a fixed energy grid. The spectrum may be read as a two-column (wavelength, flux) file — for example a MUSCLES spectral energy distribution scaled to the orbital distance ([K. France et al. 2016](#)) — or represented as a broken power law over an EUV band and an X-ray band normalized to the band luminosities. Because the one-dimensional model is spherically symmetric and cannot resolve the day–night illumination geometry, the ionizing flux entering the hydrodynamics is reduced by a redistribution factor (the “Rate/2 + Mdot/2” approximation), valid as an average over the irradiated hemisphere; diagnostics that model the one-sided illumination explicitly (e.g. the Ly α transfer of Section 2.5) instead use the full incident flux, to avoid double-counting the geometry.

The photoionization cross sections combine exact and fitted forms. Hydrogen and He II use the analytic hydrogenic cross section with the Gaunt-factor correction, exact at threshold ($6.3/Z^2 \text{ Mb}$). The He I ground state uses the single-shell analytic fit of [D. A. Verner et al. \(1996, VFKY96\)](#) by default, which agrees with the legacy two-term **ATES** fit to within 14% over 24.6–500 eV; the metals and He II use the same VFKY96 functional form. The metastable He 2^3S level — the

lower state of the 10830 triplet, whose threshold lies at 4.77 eV in the mid-ultraviolet — is not in the Verner tabulation, so it is represented by two VFKY96 terms (the $2s$ -shell channel and the resonance-averaged He II $n = 2$ channel) joined by a log-linear bridge across the Cooper minimum. This construction reproduces the Norcross (1971) tabulation shipped with **p-winds** ([L. A. dos Santos et al. 2022](#)) to within $\sim 10\%$ and an independent TOPbase/Opacity Project R-matrix cross section to within $\sim 8\%$ in the smooth near-threshold region. The autoionizing resonances that TOPbase resolves above $\sim 35 \text{ eV}$, and the smooth high-energy tail above the fitted cutoff, each change the He 2^3S photoionization rate by only 1–2%; because these are below the fit and composition uncertainties, they do not affect the 10830 prediction (a documented approximation valid for the smooth resonance-averaged background, which is what the broadband rate integral samples).

The energy deposited per photoionization is the photoelectron energy above threshold, summed over every absorber the ionization network carries: H I, He I, He II, the metastable He 2^3S , and the photoionizable trace metals (C, N, O, Mg, Ca, Na, and Fe ions); molecular runs add an H₂ photoionization term. The photoelectric heating efficiency is not unity: a fraction of each primary photoelectron’s energy drives secondary ionizations rather than heating the gas, and **EXHALE** reduces the deposited heat by that fraction following the ionized-fraction partition of Shull & van Steenberg (1985), applied as an optional, staged activation, rather than treating the secondary ionizations as a separate channel.

Beyond the direct photoelectric heating, **EXHALE** tracks four reprocessing channels that return trapped line and recombination energy to the gas. Balmer-continuum photoelectric heating draws on the excited $n = 2$ hydrogen population ([D. Christie et al. 2013](#); [C. Huang et al. 2017](#)). Ly α de-excitation heating returns the 10.2 eV excitation energy of Ly α -pumped $n = 2$ hydrogen as heat when the level is collisionally de-excited, in the trapped-Ly α picture of [C. Huang et al. \(2023\)](#); the pumping field is the in-line Ly α transfer of Section 2.5. Helium recombination-coupling heating captures helium recombination photons that photoionize and heat hydrogen. Finally, He 2^3S +H Penning-ionization heating uses the temperature-dependent rate of [A. R. Taylor et al. \(2025\)](#), with a corresponding He 2^3S +H₂ term in molecular runs. The code writes each of these heating terms, together with the direct photoionization heating of every absorber, as a separate column of a diagnostic **Heating_breakdown.txt** file whose channel sum reproduces the total heating that enters the hydrodynamics (Table 1).

2.3. Coupled ionization and thermochemistry

The ionization state is computed in photoionization equilibrium as a single coupled nonlinear algebraic system that includes hydrogen, helium, and the trace metals C, N, O, Mg, Ca, Na, and Fe, each carried through its relevant ionization stages, together with the metastable $\text{He } 2^3S$ level and the excited $n = 2$ hydrogen population. Solving all species simultaneously (rather than element by element) captures the shared free-electron density and the charge-exchange coupling between hydrogen and the metals. The system is solved with an analytic-Jacobian Newton iteration and a MINPACK `hybrd1` dogleg fallback for the cells where the Newton step leaves the physical region; the analytic Jacobian carries the great majority of cells and the fallback guarantees robustness.

The reaction rates are drawn from standard atomic-data sources, not fitted internally: radiative and dielectronic recombination from the Badnell data (N. R. Badnell 2006; N. R. Badnell et al. 2003) (with Storey & Hummer hydrogenic recombination where appropriate; P. J. Storey & D. G. Hummer 1995), collisional ionization from the G. S. Voronov (1997) fits, and charge exchange between hydrogen and the metals from J. B. Kingdon & G. J. Ferland (1996), taken in the tabulation of C. Huang et al. (2023). The trace-metal abundances are set at runtime through a configuration file, so a model can be run with or without metals without recompiling. Because the metals contribute to the mass, electron, and particle budgets, they are included self-consistently in the bulk-gas equation of state, so their presence feeds back on the base density, pressure, and mean molecular weight rather than being a passive overlay.

One entry of that tabulation is corrected here. As printed, the two oxygen rows of Table 4 of C. Huang et al. (2023) are inconsistent with detailed balance: the $\exp(-227/T)$ Boltzmann factor sits on the $\text{O}^+ + \text{H}$ row, whereas $\text{IP}(\text{O I}) = 13.6181$ eV exceeds $\text{IP}(\text{H I}) = 13.5984$ eV, so that $\text{O} + \text{H}^+ \rightarrow \text{O}^+ + \text{H}$ is the endothermic direction ($\Delta E/k = 227.7$ K, matching the printed argument to 0.3%) and is the one that must carry it. Read as printed the pair violates detailed balance by a factor of 1.47 at 8000 K; exchanging the two reactant labels restores it to within 9%. Both the expressions of C. Huang et al. (2023) and the two oxygen fits carried by CLOUDY (C. M. Gunasekera et al. 2025) descend from the same calculation (P. C. Stancil et al. 1999), and the two sources attach the labels the opposite way, agreeing to 1–3% per direction under the exchange; this is therefore a check of the assignment and not an independent determination of the rate. Helium is the only

other row of Table 4 carrying an explicit Boltzmann factor whose placement is fixed by the ionization potentials, and it is printed correctly, its argument 12.75 (in units of 10^4 K) matching $\text{IP}(\text{He I}) - \text{IP}(\text{H I}) = 12.752$. The most likely reading is therefore a transposition of the two oxygen labels rather than an error in the coefficients, and we adopt the exchanged assignment. This concerns the printed table only; we make no claim about the coefficients used in the calculations of C. Huang et al. (2023). In our WASP-121 b model the correction leaves the wind essentially unchanged — the mass-loss rate shifts by 0.05% and the temperature by less than 10 K — but it raises the neutral-oxygen density in the ionized wind by a factor of 1.6, which would matter for any O I line prediction.

2.4. Cooling

The gas loses energy through several radiative channels, each of which EXHALE writes as a separate column of a diagnostic `Cooling_breakdown.txt` file whose channel sum reproduces the total radiative cooling. Collisional (line) excitation cooling is split by absorber into an H I term — dominated by $\text{Ly}\alpha$ and evaluated with the CHIANTI-consistent form of R. Cen (1992), consistent with J. H. Black (1981) — and separate He I and He II terms. Collisional-ionization cooling removes the ionization potential in each ionizing collision of H I, He I, and He II. Radiative recombination cooling of H II, He II, and He III uses case-B rates (Hui & Gnedin 1997). Free-free (bremsstrahlung) emission scales as $T^{0.55} n_e n_{\text{ion}}$ and sums the contributions of hydrogen, helium, and the metal ions, using the free-free Gaunt factor of van Hoof et al. The metastable helium level contributes a $2^3S \rightarrow 2^3P$ 10830 collisional-excitation cooling term. Metal-line emission is treated as described next.

Metal-line cooling uses closed-form analytic coefficients fitted to the CHIANTI database (K. P. Dere et al. 1997; G. Del Zanna et al. 2021) rather than large runtime tables. A single-channel resonance line follows the exact two-level expression

$$\Lambda(T) = \frac{8.629 \times 10^{-6}}{g_l \sqrt{T}} \Upsilon(T) \Delta E e^{-\Delta E/k_B T}, \quad (2)$$

with the slowly varying effective collision strength $\Upsilon(T)$ fitted in the Burgess–Tully form; this reproduces the Mg I, Mg II, Ca II, and Na I D cooling to a few percent. Species with many contributing transitions — Fe II and the C/N/O ions — are represented by a sum of effective two-level groups, $\Lambda(T) = T^{-1/2} \sum_i A_i e^{-T_i/T}$, whose group temperatures map onto physically distinct transition manifolds (infrared fine structure, optical metastable, ultraviolet resonance, and a high-energy

tail). These CHIANTI fits replace an earlier set of C/N/O formulas inherited from AIOLOS that omitted nitrogen entirely and deviated for oxygen across the wind-launch temperature range; the fitted nitrogen cooling and corrected oxygen curves are genuinely new physics rather than a re-parameterization.

For the ground-term fine-structure lines ([C II] 158 μm , [O I] 63 μm) the optically thin coronal limit overestimates the cooling at the dense base, where the densities exceed the line critical densities. In that regime the fine-structure floor is replaced by the exact two-level statistical equilibrium solution, which saturates the line at high density and recovers the coronal value in the low-density limit; this also restores the hydrogen-collision de-excitation channel that the electron-only coronal curves omit. These two lines are additionally allowed to be optically thick. Their line-center optical depth is accumulated over the column between the emitting cell and the top of the domain, and the corresponding escape probability $\beta(\tau)$ — the plane-parallel Doppler form of D. Hollenbach & C. F. McKee (1979) and T. de Jong et al. (1980), renormalized by a factor two so that $\beta(0) = 1$ exactly, since radiation escaping downward is absorbed by the lower atmosphere and leaves the modeled gas either way — multiplies the spontaneous rate, $A_{ul} \rightarrow \beta A_{ul}$, inside the two-level solution. Placing β inside matters: in the subcritical limit the cooling is fixed by the collisional excitation rate and must not depend on β , which an external factor would violate. In the tenuous wind $\beta \rightarrow 1$ and the optically thin result is recovered; at the HD 189733 b base, where the layer is cold and dense, $\tau([\text{O I}] 63 \mu\text{m}) \simeq 3$ and the trapping is a factor of a few. Resonance metal lines that can be thick in a metal-rich wind (Mg II h&k) are still treated as optically thin; this is a documented limitation rather than a solved problem.

All of the CHIANTI coefficients are fitted over 10^3 – 10^5 K, and below that floor they are not merely unvalidated but unphysical: their softest exponentials carry excitation temperatures of several hundred to a few thousand kelvin that correspond to no ground-term splitting of the ion in question. The coronal part is therefore removed smoothly below 10^3 K by a cutoff factor that is exactly unity at and above the floor, with continuous value and temperature derivative (the bracketing temperature solve and the semi-implicit energy update both differentiate the cooling in T). Only the explicit two-level fine-structure terms, which are valid there, survive below the floor. The width of the cutoff is a modeling choice, not a measurement. In molecular runs an H_3^+ ro-vibrational cooling term is added (Miller et al. 2013).

The radiative channels above are the source term in the thermal energy equation; the gas also loses energy dynamically. Adiabatic (expansion) cooling — the $P dV$ work done by the expanding wind — is carried directly by the hydrodynamics rather than as a radiative loss, and it is typically the dominant energy sink in the outer wind, where it becomes comparable to or exceeds the radiative terms; the structure figures show it as a separate curve. Thermal conduction and viscous dissipation are omitted by default: the viscosity coefficient defaults to zero and thermal conduction is deferred, so neither enters the present models. Table 1 collects all of the heating, cooling, and dynamical channels together with their references.

2.5. The metastable helium triplet and in-line Ly α radiative transfer

The He I 10830 and Balmer diagnostics require the excited-state populations to be solved together with the wind. The metastable He 2^3S level is treated as a separate species in the coupled system. In steady state its abundance relative to He II is

$$\frac{n(\text{He } 2^3S)}{n_{\text{He}^+}} \simeq \frac{\alpha_3 n_e}{\Phi_3 + A_{31} + (q_{31a} + q_{31b})n_e + Q_{31}n_{\text{H}}}, \quad (3)$$

where α_3 is the recombination rate into 2^3S , Φ_3 the mid-ultraviolet photoionization rate, $A_{31} = 1.27 \times 10^{-4} \text{ s}^{-1}$ the radiative decay, and q_{31} , Q_{31} the electron and hydrogen collisional sinks. The hydrogen-collision sink includes the temperature-dependent Penning ionization rate of A. R. Taylor et al. (2025), which replaces the older constant value and is often the dominant 2^3S loss near the dense base.

The Balmer lines absorb from the $n = 2$ hydrogen population, which is built following the non-LTE rate equilibrium of D. Christie et al. (2013): collisional excitation, recombination cascade, two-photon decay, ℓ -mixing between $2s$ and $2p$, and — dominantly — Ly α radiative pumping $B_{1s \rightarrow 2p} \bar{J}_{\text{Ly}\alpha}$. Rather than a full Monte Carlo, which is intractable at the Ly α line-center optical depths of a hot Jupiter ($\tau_0 \sim 10^7$ – 10^8 , where core-skipping is both forbidden and incompatible with the core-dominated pumping), EXHALE evaluates $\bar{J}_{\text{Ly}\alpha}$ from a local escape-probability radiative-transfer field every time step. The treatment follows the Neufeld/Harrington wing-escape solution, which is accurate in the very optically thick limit $a\tau_0 \gg 1$ that applies here (D. A. Neufeld 1990). It combines an internal source term (recombination cascade plus electron-impact excitation) closed with the wing escape probability $\beta = \pi^{-1/4} \sqrt{a/\tau}$ and a $(1 - \beta)$ factor that prevents the source function from diverging as the gas ionizes in

Table 1. Heating, cooling, and dynamical energy channels in **EXHALE**. Each radiative channel is written to a separate column of **Heating_breakdown.txt** or **Cooling_breakdown.txt**, whose channel sums reproduce the total heating and cooling that enter the hydrodynamics.

Channel	Reference / form
Heating	
Photoionization, H I	photoelectron energy above threshold
Photoionization, He I, He II	
Photoionization, He 2^3S metastable	mid-UV, > 4.77 eV
Photoionization, C/N/O/Mg/Ca/Na/Fe ions	
Photoionization, H ₂	molecular runs
Secondary-ionization efficiency reduction	Shull & van Steenberg 1985
Balmer-continuum photoelectric ($n = 2$ H)	D. Christie et al. (2013) ; C. Huang et al. (2017)
Ly α de-excitation heating	C. Huang et al. (2023) ; Sec. 2.5
He recombination-coupling heating of H	
He 2^3S +H Penning ionization	A. R. Taylor et al. (2025)
He 2^3S +H ₂ Penning ionization	molecular runs
Radiative cooling	
Collisional excitation, H I (Ly α)	R. Cen (1992) ; J. H. Black (1981)
Collisional excitation, He I, He II	
Collisional ionization, H I/He I/He II	
Radiative recombination (case B)	Hui & Gnedin 1997
Free-free, H + He + metal ions	van Hoof Gaunt factor
He 2^3S metastable $2^3S \rightarrow 2^3P$ 10830	
Metal lines, C/N/O/Mg/Ca/Na/Si/K/S/Fe	K. P. Dere et al. (1997) ; G. Del Zanna et al. (2021)
H ₃ ⁺ ro-vibrational	Miller et al. 2013; molecular runs
Dynamical (energy equation)	
Adiabatic (expansion) cooling	$P dV$ work of the wind
Thermal conduction, viscous dissipation	omitted by default

the outer wind, plus a trapped stellar-beam term whose penetration through the broad stellar-line wings uses the stellar line width. An optional Sobolev (wind-velocity) escape term is included; for a hot Jupiter it changes $\bar{J}_{\text{Ly}\alpha}$ by less than a percent because the velocity gradient and the Ly α opacity scale together. For percent-level H α work under one-sided illumination, where the pumping field is only cylindrically symmetric, **EXHALE** can instead ingest a two-dimensional scattering-rate grid from the **LaRT** Monte Carlo code ([K.-i. Seon & C.-G. Kim 2020](#)).

2.6. Diffusive separation

Because helium is four times heavier than hydrogen, it is under-dragged by the escaping hydrogen wind, so the escaping He/H ratio can fall below the bulk abundance with increasing altitude. **EXHALE** retains its single bulk-momentum formulation but represents this separation with a molecular- plus eddy-diffusion term that lets the He/H ratio vary with radius, in the spirit of [A. R. Taylor et al. \(2025\)](#) and consistent with the multi-fluid result of [L. Xing et al. \(2023\)](#). The homopause where the diffusive separation sets in is controlled by an eddy-diffusion coefficient that can be supplied from the lower-atmosphere model (Section 2.7). The same framework extends to He/metal separation. This altitude-dependent compo-

sition is what the transit post-processor requires for a physically consistent He 10830 column; it is optional and off by default.

2.7. Lower-atmosphere boundary

The base of the escape model sits near a μ bar pressure level, below which the atmosphere is denser, cooler, and possibly molecular. **EXHALE** couples to this region through an optional handoff file (`base.inp`) that overrides the base temperature, base radius, He/H ratio, and eddy-diffusion coefficient before the derived constants are computed. A tiered set of lower-atmosphere models writes this file: an analytic hydrostatic column that returns the 1μ bar radius and a molecular-hydrogen fraction; a molecular-chemistry option with H_3^+ cooling for genuinely cool sub-Neptunes; and, at the most complete tier, a photochemical column from the **VULCAN** code (S.-M. Tsai et al. 2017), whose disequilibrium composition replaces the chemical-equilibrium estimate (C. Visscher et al. 2006; J. I. Moses et al. 2011). For a warm hot Jupiter the photochemical base dissociates far more hydrogen than chemical equilibrium predicts, so the atomic base that the escape model assumes rests on photochemistry rather than on thermal dissociation; **VULCAN**, however, does not release the atomic metals (Na/Mg/Ca/Fe), whose abundances remain user-supplied.

The coupling is one-way by default (lower atmosphere $\rightarrow$ base $\rightarrow$ escape). Two caveats bound its validity. First, the composition must not be double-counted: if the handoff file sets the He/H ratio from the lower model, the metal abundances should be that model’s atomic-release values rather than solar totals. Second, and most important, the handoff assumes that the base pressure lands below the thermospheric temperature rise. If it instead falls inside a thermospheric temperature inversion — where the escape model’s own heating has already raised the temperature — then adopting a cool lower-atmosphere T and composition at that level double-counts the transition region and misrepresents the base state. The handoff is therefore valid only when the chosen pressure level sits beneath the onset of thermospheric heating; for dayside-hot or strongly irradiated cases the base radius from an isothermal or radiative-equilibrium column is a lower bound that must be checked against the escape model’s own temperature structure.

3. DIFFERENCES FROM ATES

EXHALE inherits its hydrodynamic core and boundary/initial-condition skeleton from **ATES** (A. Caldiroli et al. 2021, 2022): the Godunov finite-volume

scheme, the SSPRK time integrator, the HLLC default with Roe and LLF alternatives, PLM and ESWENO3 reconstruction, the 500-cell mixed grid, the two-stage PLM $\rightarrow$ WENO3 relaxation to $\Delta\dot{M}/\dot{M} < 10^{-3}$, the fixed-reservoir lower boundary with the one-way velocity valve, the supersonic-outflow upper boundary, and the cold neutral isothermal initial condition with a linear velocity seed. The ionization is likewise solved in photoionization equilibrium with a MINPACK method, and ion advection is applied as a post-processing step. What **EXHALE** adds is in the microphysics and in the steady-state solver, not in the hydrodynamic backbone. Table 2 summarizes the differences, which are then discussed.

Solver.—**ATES** reaches steady state by time relaxation only, so its convergence is the mass-flux flatness du ; **EXHALE** keeps that path but adds a direct steady-state solver (Section 2.1) that drives the discrete residual toward zero, a stricter standard that also exposed cases where the du criterion stops at a premature, non-steady snapshot. **EXHALE** also replaces the explicit source update with a semi-implicit energy step and a bracketing (Brent) temperature solve that avoids the hot-root trap of stiff metal cooling.

Metal chemistry.—The central physics addition is the coupled trace-metal network. **ATES** carries hydrogen and helium only; **EXHALE** solves C, N, O, Mg, Ca, Na, and Fe through their ionization stages in the same nonlinear system, with Badnell radiative and dielectronic recombination, Voronov collisional ionization, and Kingdon & Ferland charge exchange, and it feeds the metals back into the equation of state. This enables both the metal-line cooling and the metal transit diagnostics that **ATES** cannot produce.

Excited states and radiative transfer.—**EXHALE** adds the He 2^3S metastable network for He 10830, the non-LTE $n = 2$ hydrogen population for the Balmer lines, and the in-line Ly α escape-probability field that pumps it — none of which are in **ATES**. These, with the diffusive He/H separation and the tiered lower-atmosphere coupling, turn the inherited escape solver into a line-diagnostic model. Notably, the base oscillation (“breathing”) phenomenology and its mitigations (the smooth valve, the advection guard) are **EXHALE**-era findings not present in the **ATES** description.

4. RESULTS

We apply **EXHALE** to four transiting planets that span the relevant parameter range and have published transit-absorption measurements. For each we report the Newton-converged wind structure (Figures 1–4), the

Table 2. Algorithm and physics: EXHALE compared with ATES v2.0

Axis	ATES v2.0	EXHALE
Hydro core	Godunov, HLLC/Roe/LLF, PLM/ESWENO3, SSPRK	Same (inherited)
Steady state	Time relaxation only (<i>du</i> criterion)	Time relaxation + JFNK Newton finish; residual-based stop
Energy update	Explicit source	Semi-implicit energy + Brent <i>T</i> -solve
Metals	None	C, N, O, Mg, Ca, Na, Fe in the coupled system, in the EOS
Ionization solve	MINPACK dogleg	Analytic-Jacobian Newton + MINPACK fallback
Recombination	Standard tables	Badnell RR + DR
Cooling	Ly α , recomb, free-free (H/He)	Adds CHIANTI metal-line fits (with fine-structure saturation)
He triplet	Not solved	He 2 ³ <i>S</i> metastable network in-loop
Balmer / Ly α RT	Not solved	<i>n</i> = 2 non-LTE + in-line Ly α escape-probability field
Charge exchange	—	Kingdon & Ferland H $\leftrightarrow$ metal
Diffusion	None (fixed He/H)	Optional He/H and He/metal diffusive separation
Lower atmosphere	Fixed base	Tiered base (analytic / molecular / VULCAN handoff)
Transit spectra	—	He 10830, Ly α , H α , H β , metal doublets

steady-state mass-loss rate, and the synthetic transit depths against observations (Figures 5–7). All four solutions are converged with the JFNK steady-state solver rather than stopped on the *du* criterion, reaching discrete steady residuals of 7.7×10^{-4} , 5.4×10^{-4} , 1.6×10^{-4} and 5.7×10^{-5} for HD 209458 b, HD 189733 b, WASP-52 b and WASP-121 b; this matters quantitatively, because a *du*-threshold stop alone carries a path-dependent spread of several percent in $\dot{M}$.

Quoted transit depths are the line-center excess absorption of the model curve, with the instrument-convolved value given after a slash where the convolution changes it appreciably. The figures additionally include the broadening by the (tidally locked) planet rotation, which lowers the plotted peak further; the planet, stellar, and spin parameters all come from each run directory’s own input file. Of the four models only HD 189733 b carries the diffusive He/H and He/metal separation of Section 2.6; the other three are run with a fixed He/H ratio, so composition gradients are not a differentiating factor between them. The structure figures resolve both the radiative cooling and the heating into their individual channels, with the collisional-excitation cooling split by absorber into an H I (Ly α -dominated) term and the helium terms.

4.1. HD 209458 b

EXHALE relaxes HD 209458 b to a steady transonic wind with a mass-loss rate of $\log_{10}(\dot{M}/\text{g s}^{-1}) = 9.31$. The temperature climbs from the ~ 1450 K base through a steep thermosphere to a broad maximum of $\approx 7.8 \times 10^3$ K near 1.4 – $1.6 R_p$, and the gas becomes predominantly ionized above $\sim 1.1 R_p$, where H II overtakes H I and helium is mostly singly ionized (Figure 1). The outflow is slow and subsonic across the modeled thermosphere, reaching 0.7 km s^{-1} at the $2 R_p$ escape radius and crossing the sonic point only at $\approx 4.1 R_p$, well outside it. In the cooling budget the Ly α -dominated collisional excitation and recombination channels balance the photoionization heating in the ionized layer, while adiabatic expansion becomes comparable to the radiative losses in the outer wind.

The synthetic He 10830 triplet has a line-center depth of 9.7/9.1% and Ly α reaches 25.1%, while H α is weak at 0.22% (Figures 5, 6, and 7). The metal resonance lines of this solar-composition run are predicted at 10.3/5.9% (Mg II h&k), 3.1% (Ca II H&K), and 1.4/1.3% (Na I D); the large convolution effect on Mg II reflects how narrow its saturated core is. The digitized wing Ly α profile of A. Vidal-Madjar et al. (2003) is of comparable magnitude to the model, although the Ly α core is inaccessible

ble to observation because of interstellar and geocoronal absorption. The $H\alpha$ trace of A. G. Jensen et al. (2012) scatters by ± 0.7 – 0.9% about zero, several times the model depth, so it neither confirms nor excludes a signal this weak; the model line is also narrower than the broad, antisymmetric structure in the digitized data.

4.2. HD 189733 b

For the denser, more strongly bound HD 189733 b the model gives a mass-loss rate of $\log_{10}(\dot{M}/\text{g s}^{-1}) = 9.04$, the lowest of the four. The thermosphere is hotter and more compact than that of HD 209458 b, with a temperature maximum of $\approx 1.25 \times 10^4$ K, and the wind is again slow and subsonic through the modeled region, reaching only 0.5 km s^{-1} at $2 R_p$ (Figure 2). Hydrogen and helium ionize over a narrow shell just above the base; the diffusive He/H separation included in this run lets the He/H ratio fall with altitude.

This is the planet for which the base grid has to be refined. On the default spacing its base density scale height spans only ~ 27 cells, the least margin of the four, and the resulting stationary cell-to-cell entropy mode described in Section 2.1 survives in the converged state. The model quoted here therefore doubles the base resolution, which removes the mode from the interior cells; a residual alternating signature remains in the first two cells above the boundary, where it measures the ghost-to-cell density jump rather than the entropy mode, and is a documented numerical artifact of the lower boundary rather than a physical feature. The base temperature this run settles to, ~ 490 K, is set by where the trapped fine-structure cooling of Section 2.4 balances the local metal photoionization heating; the model contains no stellar optical/infrared absorption and no thermal background, so nothing enforces a floor at the equilibrium temperature for a layer that has shielded out the XUV.

The synthetic He 10830 depth is 2.5% and $H\alpha$ is 0.52% (Figures 5 and 6). The model He 10830 is about three times deeper than the M. Salz et al. (2018) detection, which peaks near 0.9% , while sharing the double-peaked triplet shape. Conversely, the clean $\approx 2.2\%$ $H\alpha$ dip digitized from A. G. Jensen et al. (2012) is deeper than the model $H\alpha$, indicating that the $n = 2$ population and its $\text{Ly}\alpha$ pumping in this run under-produce the observed Balmer absorption. The opposite sign of these two mismatches suggests the discrepancy lies in the excited-state microphysics rather than in the overall wind density.

4.3. WASP-52 b

WASP-52 b is a low-gravity, strongly irradiated planet, and the model produces a correspondingly large

mass-loss rate of $\log_{10}(\dot{M}/\text{g s}^{-1}) = 11.61$. The temperature reaches $\approx 9.1 \times 10^3$ K and the gas is highly ionized through the extended thermosphere, which is modeled here on a domain reaching to $10 R_p$; the wind accelerates smoothly, passes 10 km s^{-1} at the $2 R_p$ escape radius, and crosses the sonic point just outside it at $\approx 2.2 R_p$ (Figure 3). The low assumed He/H ratio ($\simeq 0.02$) still leaves a large neutral-hydrogen and metastable-helium column along the extended line of sight.

The resulting transit depths are the largest of the sample: the model He 10830 triplet reaches $23.5/22.9\%$, $H\alpha$ 3.5% , and $\text{Ly}\alpha$ is fully saturated near 100% (Figures 5–7). The metal lines are deeper still, with Mg II at $65.0/57.2\%$ and Ca II at 14.8% . These greatly exceed the measured absorption — the J. Kirk et al. (2022) He 10830 spectrum peaks near 3.6% and the G. Chen et al. (2020) $H\alpha$ spectrum is consistent with a sub-percent signal. The model therefore over-predicts the WASP-52 b line depths by a factor of several. The most likely contributors are the large radial extent of the absorbing column on the $10 R_p$ domain and the assumed composition; a reduced He/H ratio and a line of sight truncated nearer the Roche lobe would both lower the predicted depths, and quantifying this sensitivity is left to future work rather than tuned here. A companion analysis of the same target separates the two levers. Holding the microphysics and composition fixed and truncating the line-of-sight integration at the L_1 Roche point, rather than carrying it to $10 R_p$, lowers the synthetic He 10830 depth by a factor of 3–10 on its own, because most of the absorbing column on the extended domain lies beyond the radius the planet can gravitationally retain. Reducing the helium abundance from solar to $\text{H}/\text{He} \simeq 98/2$ then accounts for a further factor of about two, and the two together reproduce the measured depth. The larger of the two effects is therefore the domain, which argues that the over-prediction reported here is primarily a geometry problem, to be addressed with a Roche-lobe truncation of the transit integration rather than by adjusting the microphysics.

4.4. WASP-121 b

WASP-121 b is the most extreme target, an ultra-hot Jupiter close to Roche-lobe overflow. The model, computed without diffusive separation, gives a mass-loss rate of $\log_{10}(\dot{M}/\text{g s}^{-1}) = 13.17$, the highest of the sample. The thermosphere is very hot — of order 10^4 K through most of the modeled shell and rising steeply toward the $1.5 R_p$ escape radius — and the wind reaches 14 km s^{-1} there, crossing the sonic point at $\approx 1.6 R_p$, just beyond the escape radius (Figure 4). The gas is strongly ionized, with metal ions such as C II, O II, and N II carrying

HD 209458 b – radial structure

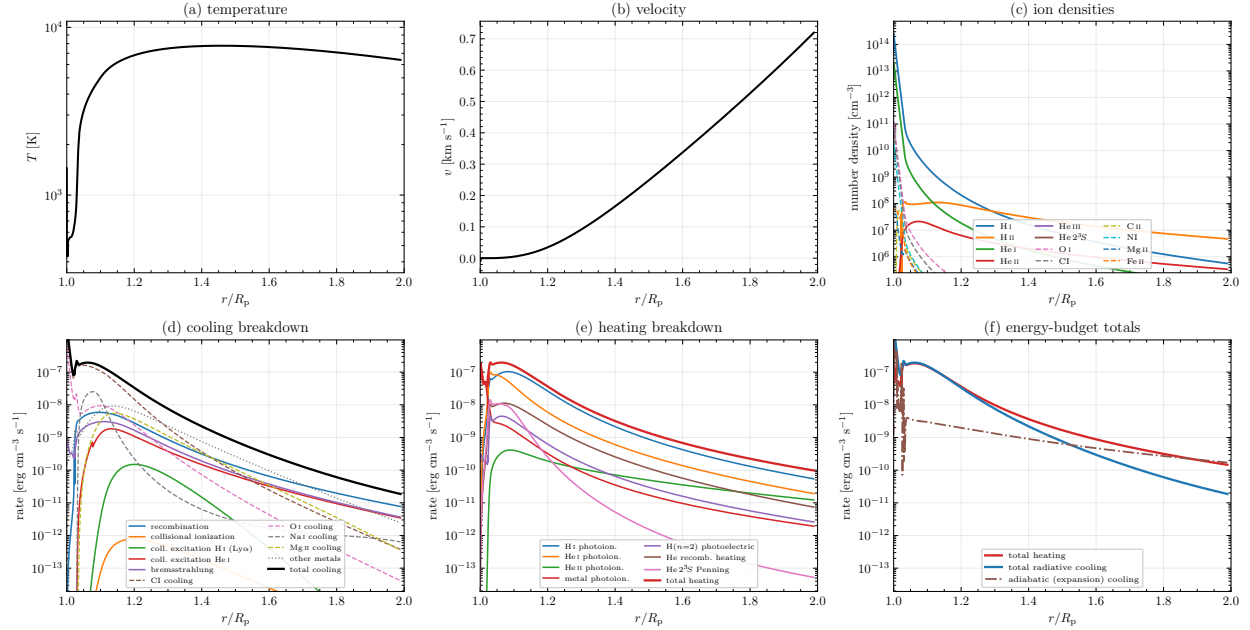


Figure 1. Radial structure of the converged HD 209458 b model: (a) temperature, (b) radial velocity, (c) ion number densities (H and He stages, the metastable He 2^3S , and the most abundant metal ions in the wind), (d) the radiative cooling breakdown, (e) the heating breakdown, and (f) the energy-budget totals. Panel (d) shows each radiative cooling channel from the model output, with the collisional-excitation cooling split by absorber into H I (the Ly α -dominated H-line cooling), He I, and He II. Panel (e) shows the photoionization heating of each absorber (H I, He I, He II, He 2^3S , and the metals), the excited-hydrogen photoelectric and Ly α de-excitation heating, and the He-recombination and He 2^3S Penning heating. Panel (f) compares the total heating, the total radiative cooling, and the adiabatic (expansion) cooling computed from the velocity field. The individual channels in panels (d)–(f) are the ones tabulated in Table 1. Curves are drawn to the $2 R_p$ escape radius. Generated by `python/make_struct_figures.py`.

an appreciable fraction of the electrons and contributing to the metal-line cooling. Because the escape radius lies inside the Roche geometry, the profile near the outer boundary is sensitive to the domain truncation, and the solution shows a boundary-layer temperature feature there.

The synthetic depths are the most nearly uniform across lines of the four planets, and several of them sit at the same value: He 10830 is 3.7%, H α 3.3%, Ly α 3.7%, Mg II 3.8%, and Ca II 3.7% (Figures 5–7). That common ceiling is geometric rather than physical. A line whose absorbing column is optically thick all the way to the edge of the modeled profile blocks the whole annulus between the planet and that edge, so its depth cannot exceed $(R_p/R_\star)^2 (r_{\text{out}}^2 - 1)$, which is 3.9% here for a profile ending at $r_{\text{out}} = 1.61 R_p$ on a $1.458 R_\odot$ star. The saturated lines sit just under that bound. Every one of these depths is therefore a lower bound imposed by the domain rather than a prediction, and none of them should be read as a measurement of how much absorption the wind actually produces: what the model establishes on this planet is that the wind is optically

thick in all five lines out to the truncation radius. A quantitative depth requires a domain carried past the Roche lobe, which is where the one-dimensional spherical geometry stops being the right description in any case. For the same reason, the fact that the He 10830 depth exceeds the S. Czesla et al. (2024) measurement, which peaks near 2%, is not yet evidence of a discrepancy in the microphysics; we note only that the observed absorption is offset in wavelength relative to the model line center. That the metal lines saturate at all is consistent with the metal-rich, highly ionized wind expected for this planet (C. Huang et al. 2023; T. T. Koskinen et al. 2022a).

5. SUMMARY

We have described EXHALE, a one-dimensional multi-frequency photoionization–hydrodynamics model for escaping exoplanet atmospheres. Building on the ATES Godunov hydro core, EXHALE adds a coupled H/He/metal ionization network with Badnell recombination, Voronov collisional ionization, and Kingdon & Ferland charge exchange; CHIANTI metal-

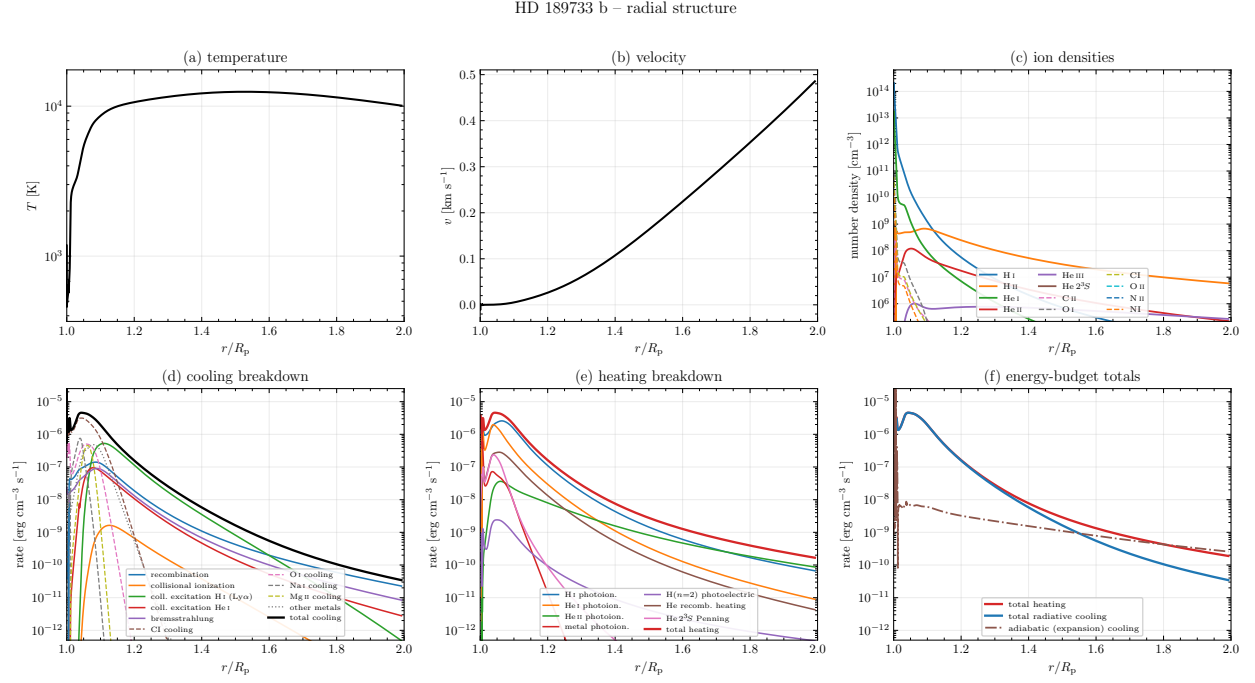


Figure 2. Radial structure of the converged HD 189733 b model, as in Figure 1. This run includes the diffusive He/H and He/metal separation. Generated by `python/make_struct_figures.py`.

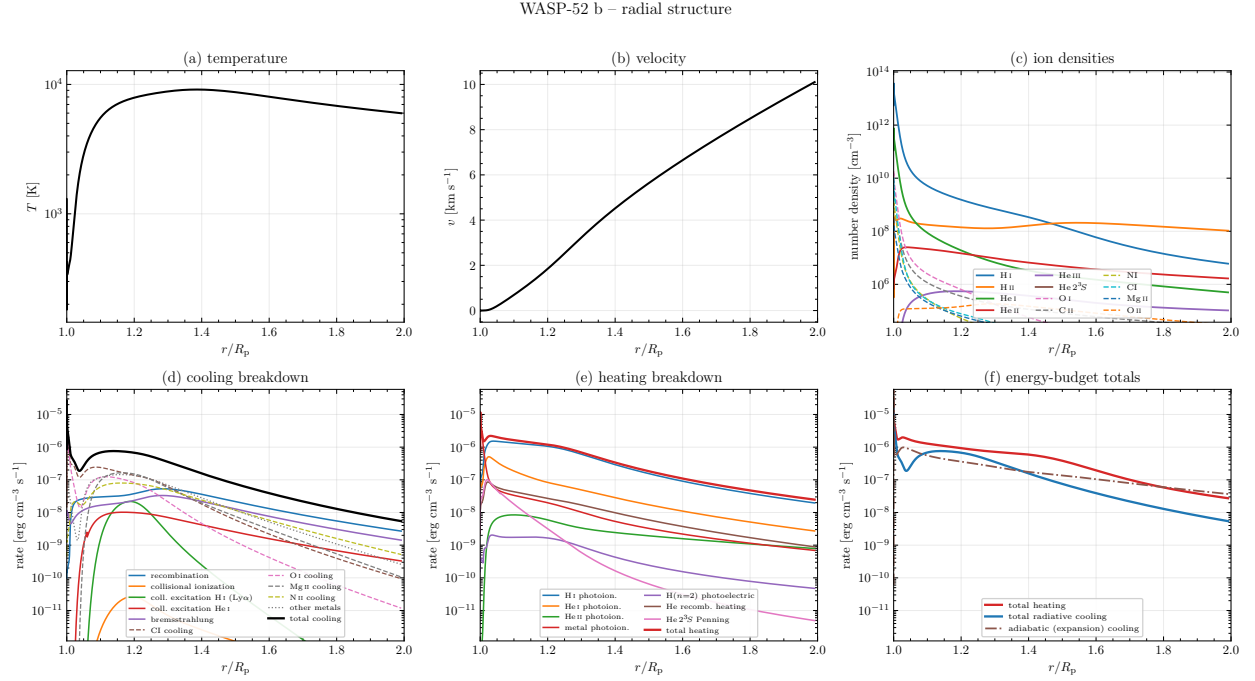


Figure 3. Radial structure of the converged WASP-52 b model, as in Figure 1, shown to the $2 R_p$ escape radius (the computational domain extends to $10 R_p$). Generated by `python/make_struct_figures.py`.

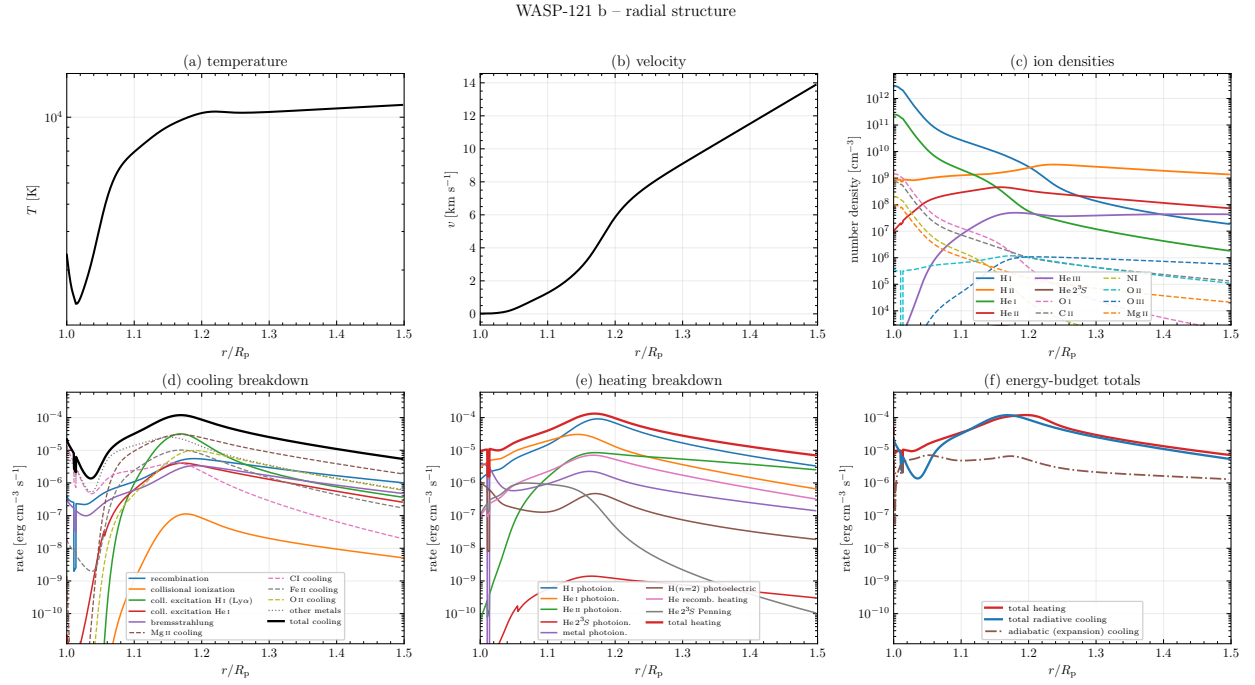


Figure 4. Radial structure of the WASP-121 b model (no diffusive separation), as in Figure 1, shown to the $1.5 R_p$ escape radius. The temperature panel is clipped to a robust upper percentile so the boundary-layer feature near the escape radius does not dominate the scale. Generated by `python/make_struct_figures.py`.

Transit transmission: He I 10830 Å

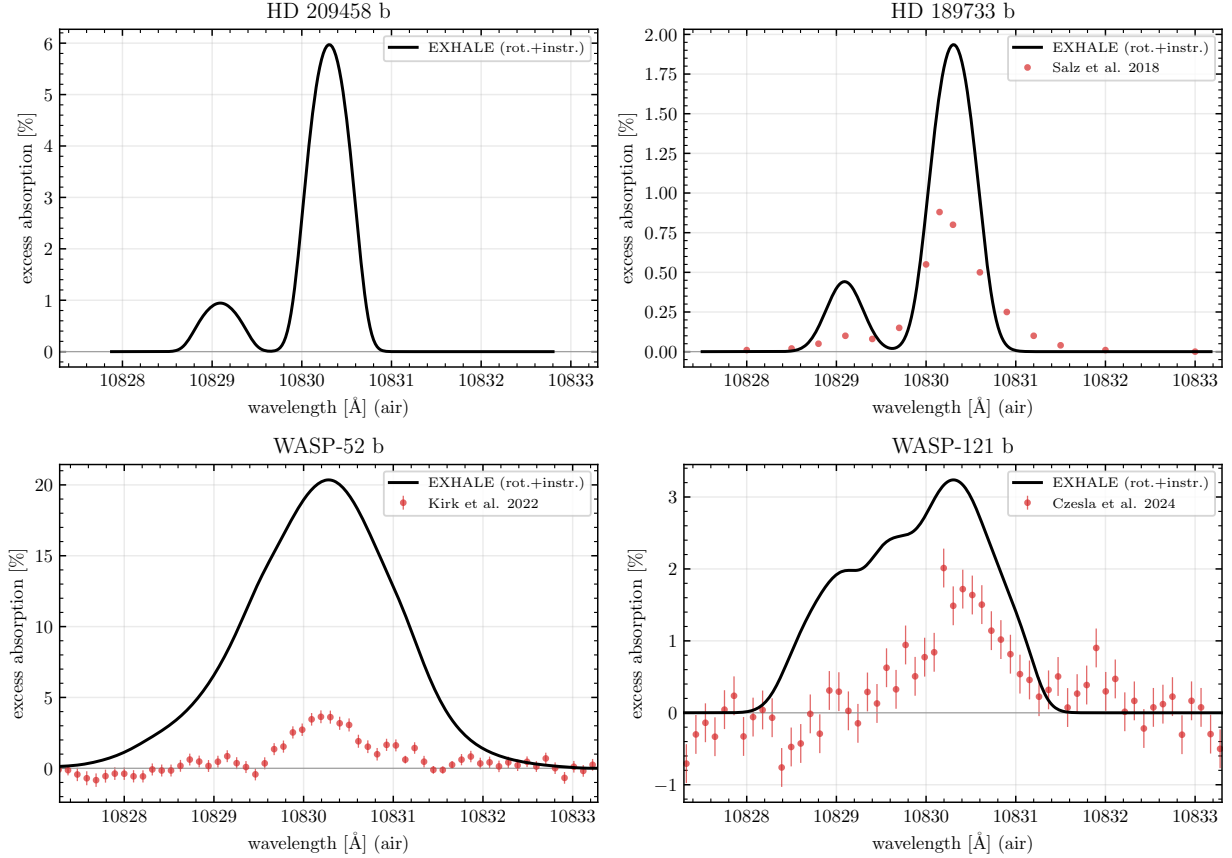


Figure 5. Synthetic He I 10830 transit spectra (rotation- and instrument-convolved) for the four planets, compared with published observations where available: M. Salz et al. (2018) for HD 189733 b, J. Kirk et al. (2022) for WASP-52 b, and S. Czesla et al. (2024) for WASP-121 b. All curves are shown as excess absorption, $(1 - T)$ in percent; each observation is converted to this common axis from its native convention (dF/F, excess-absorption percent, or normalized in/out flux). Generated by `python/make_transit_figures.py`.

line cooling; the He 2^3S metastable triplet; the non-LTE $n = 2$ hydrogen population with an in-line Ly α escape-probability field; optional diffusive He/H separation; a direct JFKN steady-state solver; and a tiered lower-atmosphere coupling. Applied to HD 209458 b, HD 189733 b, WASP-52 b, and WASP-121 b, the model produces mass-loss rates and synthetic He 10830, H α , Ly α , and metal-line transit spectra for direct comparison with observations. Across the four benchmarks the model gives mass-loss rates of $\log_{10}(\dot{M}/\text{g s}^{-1}) = 9.31$ (HD 209458 b), 9.04 (HD 189733 b), 11.61 (WASP-52 b), and 13.17 (WASP-121 b), each converged with the steady-state solver rather than stopped on the mass-flux criterion, and synthetic line depths that match the order of magnitude of the measured He 10830, H α , and Ly α absorption on HD 209458 b and HD 189733 b, while over-predicting the WASP-52 b depths and saturating

against the domain truncation on WASP-121 b. The priorities these point to are the composition and the line-of-sight domain, a transit geometry carried past the Roche lobe for the most extended winds, and the excited-state microphysics behind the opposite-sign HD 189733 b He 10830 and H α mismatches.

ACKNOWLEDGMENTS

Software: EXHALE, ATES (A. Caldiroli et al. 2021), VULCAN (S.-M. Tsai et al. 2017), LaRT (K.-i. Seon & C.-G. Kim 2020), p-winds (L. A. dos Santos et al. 2022), CHIANTI (K. P. Dere et al. 1997; G. Del Zanna et al. 2021)

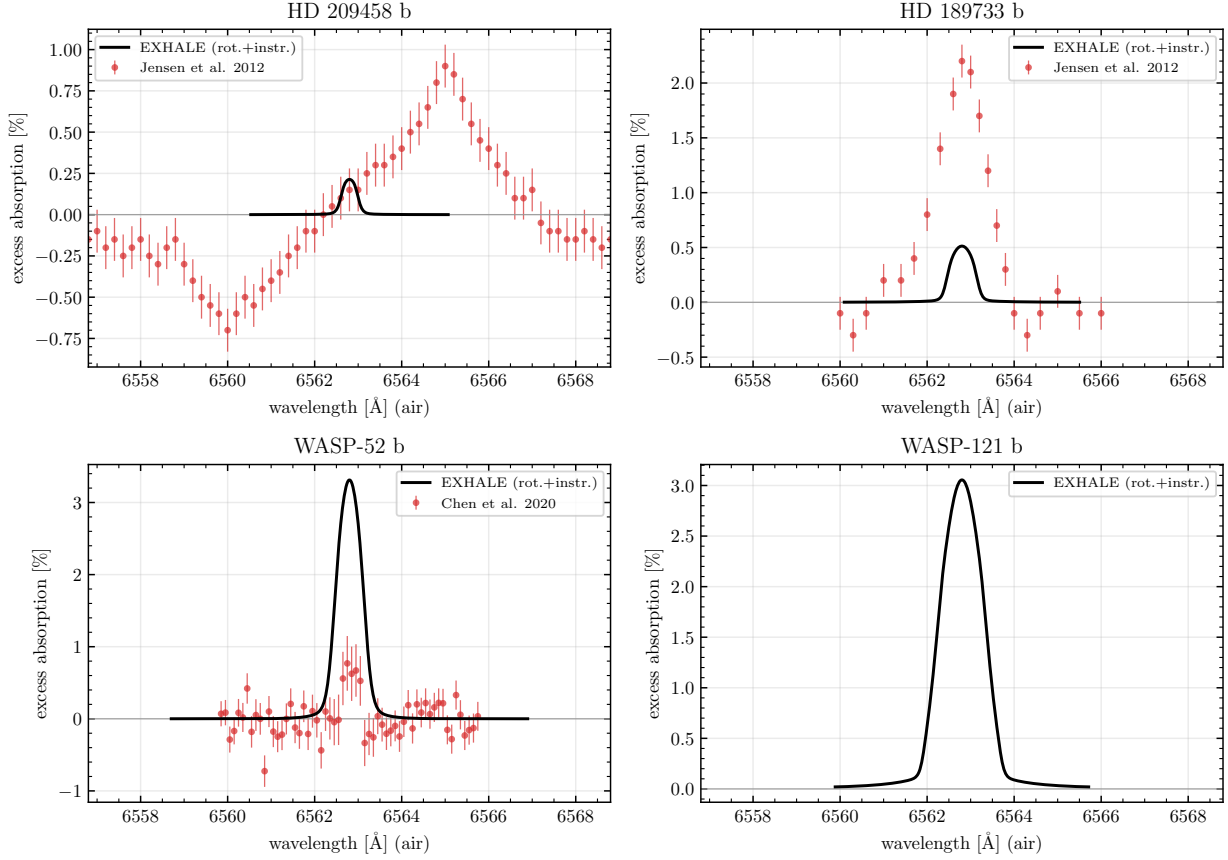
Transit transmission: $H\alpha$ 6562.8 Å

Figure 6. Synthetic $H\alpha$ transit spectra, as in Figure 5, compared with the digitized [A. G. Jensen et al. \(2012\)](#) traces (HD 209458 b and HD 189733 b) and the [G. Chen et al. \(2020\)](#) ESPRESSO spectrum (WASP-52 b). Generated by `python/make_transit_figures.py`.

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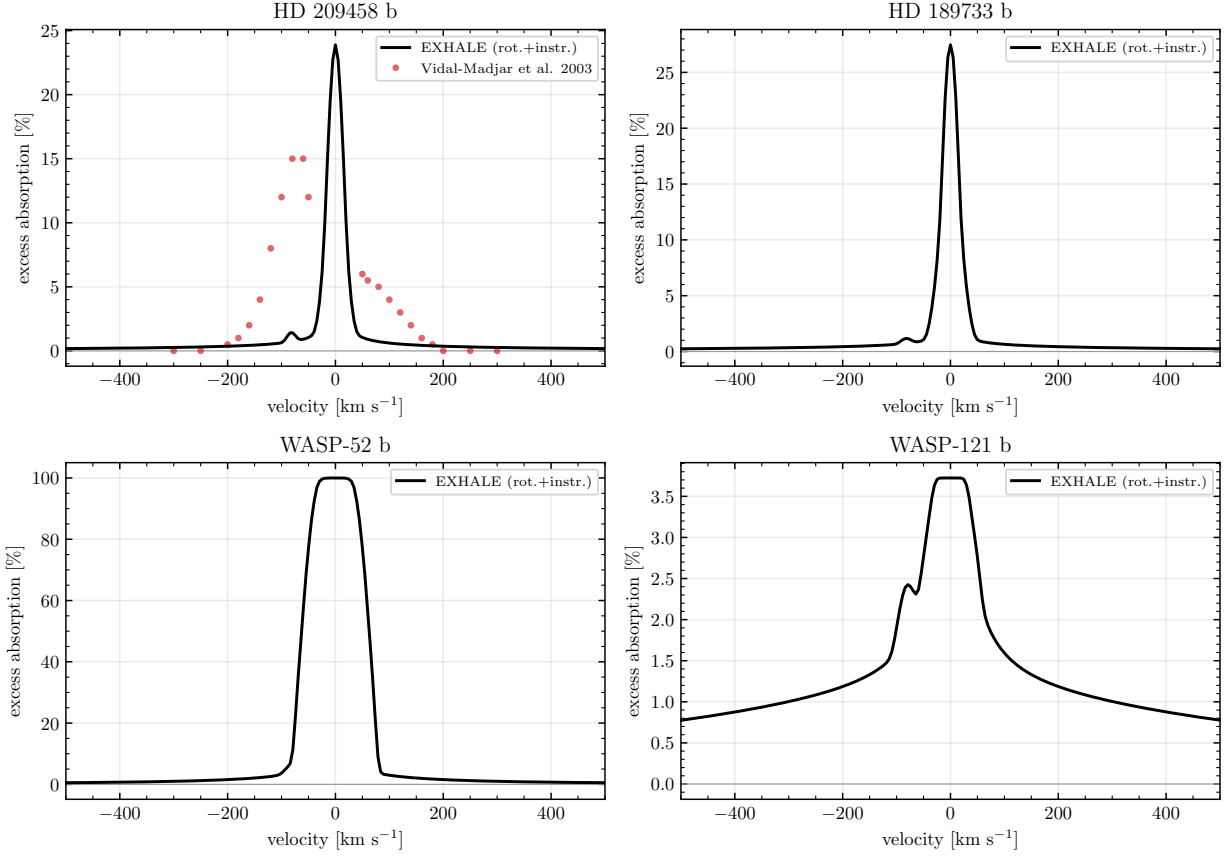
Transit transmission: $\text{Ly}\alpha$ 1215.67 Å

Figure 7. Synthetic $\text{Ly}\alpha$ transit spectra on a velocity axis, as in Figure 5. The HD 209458 b panel overplots the wing profile of A. Vidal-Madjar et al. (2003); the line core (roughly $\pm 40 \text{ km s}^{-1}$) is omitted in the data because of interstellar and geocoronal absorption. No $\text{Ly}\alpha$ measurements are available for the other three planets. Generated by `python/make_transit_figures.py`.

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